SUSHRUT MHETRAS

Windows Software Engineer

sushrutmhetras@gmail.com | +91-8857066427 | Solapur, Maharashtra

linkedin.com/in/sushrut-mhetras | github.com/sushrut007 | sushrut007.github.io/Sushrut

SUMMARY

Windows Software Engineer specializing in C++17, Qt 6.8.3 (Widgets), and C#/.NET — engineering 6 commercial Windows desktop applications at WebTech Developers Pvt Ltd. Designed software architecture for two shared cross-product C++ libraries, integrated offline AI speech recognition, and consistently delivered zero-defect test suites across AI, computer vision, document processing, networking, and licensing systems.

EXPERIENCE

Associate Windows Developer — WebTech Developers Pvt Ltd., Pune, Maharashtra / *Feb 2025 – Present*

- **Engineered 6 enterprise Windows apps** — Dictation Pro, Document Translator, My Screen Recorder, Security Monitor Pro, Text Speaker Live & WebCam Monitor — using C++17 and Qt 6.8.3 Widgets.
- **DS Text Editor (shared library):** Built a Microsoft-grade rich-text DOCX editor from scratch (QuaZIP + PugiXML); preserves 98% of formatting at <2 MB; shared submodule across 3 products.
- **DS Controls (shared library):** Architected cross-product C++ library built with CMake delivering licensing, trial management, crash reporting, encrypted WebSocket communication & auto-updates for all 7 DeskShare products.
- **AI Dictation (Dictation Pro):** Integrated OpenAI Whisper.cpp for fully offline AI speech — smallest model with industry-best accuracy/speed ratio; built 6-language post-processor (smart capitalization, punctuation, abbreviations).
- **Browser Extension:** Developed Chrome, Edge & Firefox Native Messaging extension enabling real-time dictation text injection into any browser field from the desktop application.
- **Computer Vision (Security Monitor Pro):** Implemented OpenCV multi-camera motion detection with automated scheduling, snapshot email alerts & FFmpeg H.265 recording.
- **Recovery & Media:** Multi-monitor screen recording with Recovery Management (My Screen Recorder); AI reading-plan pipeline with word-quota management (Text Speaker Live).
- **Test Quality:** Maintained 4,370+ zero-failure tests (Dictation Pro) & 3,200+ (WebCam Monitor); applied mutation testing — 89 back-tested defects — and CI-gated code review.

SKILLS

Languages	C++17, C#, Python, Java, SQL (PostgreSQL), JavaScript, HTML/CSS
Frameworks	Qt 6.8.3, Qt Widgets, .NET Framework, Flask, FastAPI, React, Node.js
Libraries & APIs	OpenAI Whisper.cpp, OpenCV, QuaZIP, PugiXML, WebSocket, WebRTC, libdatachannel, Azure Cognitive Services, OpenWeather API
Tools	Qt Creator, CMake, Git, VS Code, Jira, Salesforce, GCP, AWS
Concepts	Object-Oriented Programming (OOP), Software Architecture, Cross-platform Development, Desktop Application Development

PROJECTS

Darpan — P2P Remote Desktop & File Transfer | C++17, Qt 6.8.3, WebRTC, libdatachannel, Python / *2026*

C++17/Qt 6.8.3 peer-to-peer remote desktop and file-transfer app (Windows & Android): WebRTC DataChannels carry screen, input, and file data directly between peers; Python-only signaling server relays SDP/ICE; PIN-authenticated rooms with host-controlled permission grant/revoke.

Spam Website Detection Chrome Extension | Python, ML (SVM, Neural Networks, Random Forest), JavaScript / *Mar – May 2024*

ML-powered extension classifying and blocking malicious URLs in real time, enhancing browser security through automated threat detection.

Advanced Weather Forecasting Application | C#, .NET, OpenWeather API / *Jan 2025*

Real-time weather app with monthly forecast tracking; modular architecture, responsive UI & seamless API integration.

EDUCATION

B.Tech Computer Science & Engineering — N.K. Orchid College of Engineering & Technology, Solapur / *Nov 2021 – Feb 2025* **| CGPA: 8.65/10**

Senior Secondary (XII) — Science (State Board) — Sangameshwar College, Solapur / *Jun 2019 – May 2021* **| 93.6%**

Secondary (X) — State Board — Valentine Circle English Medium School, Solapur / *Jun 2018 – Apr 2019* **| 80.0%**

CERTIFICATIONS

- **Salesforce Developer Virtual Internship** — Salesforce, Sep 2024 / *Apex Programming, Salesforce Lightning*
- **Zensar Virtual Internship** — Zensar Technologies, Mar 2024 / *Java, Python, PL/SQL, OOP*
- **Microsoft Certified: Azure Fundamentals** — Microsoft, Sep 2022 / *Azure Cloud, IAM, VM, Cloud Storage*
- **Google Cloud Certified – Professional Cloud Architect** — Google Cloud Skills Boost / *GCP, Cloud Architecture*

AWARDS & ACHIEVEMENTS

- Winner — Hackathon with ₹10,000 cash prize
- Technical Lead & Core Team Member — Google Developer Student Club, NKOCET
- Event Manager — Computer Science Engineering Students' Association
- Conducted multiple technical workshops and sessions at home and external institutions